

BILL HUANG

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EDUCATION

University of Cambridge, MSci

2026/10-2027/07

Biochemistry

University of Cambridge, BA

2023/10-2026/07

Natural Sciences

Biochemistry, Mathematical Biology, Cell Biology, Physiology, Chemistry, Pharmacology

ACADEMIC PROJECTS

Researcher | Prywes Lab, University of Cambridge

2026/01–Present

“Engineering Alternative Carbon Fixation Strategies (HuBisCO/HuBP) in Calvin Benson Bassham (CBB) cycle”

- Engineered alternative CBB cycle to reduce RuBisCO oxygenation penalty under CO₂/O₂ trade-off constraints
- Expressed and optimised WT and 3 *R. rubrum* RuBisCO mutants, and among the 1st to express YckF in this system, achieving mg-scale purification yields
- Developed coupled NADH-linked assays by microplate spectrophotometry and NMR to quantify activity toward a novel 6C substrate (HuBP/H6P)
- Performed structure-guided mutagenesis and docking with kinetic analysis across 6 variants to evaluate ligand binding and active-site constraints, informing feasibility of benign oxygenation strategies and future substrate engineering

Research Intern | Leung Research Group, University of Melbourne

2025/07–2025/09

“Adjuvant Discovery and Mechanistic Studies to Overcome MCR-1 Mediated Colistin Resistance”

- Quantified antibacterial activity of natural adjuvants via MIC assays, identifying honokiol and pterostilbene (MIC~64 µg/mL) as active candidates
- Evaluated antibiotic synergy using FIC assays, demonstrating up to 16-fold reduction in colistin MIC (honokiol, 8 mg/mL) and 4-fold for osthonol and pterostilbene
- Established and optimised reproducible permeability (PI, NPN) assays’ workflow through systematic protocol refinement

Research Intern | CEI, University of Portsmouth

2025/06–2025/07

“Expression Optimisation and Characterisation of PET-Hydrolysing Enzymes”

- Optimised recombinant expression of PET hydrolases (CCH11, TurboPETase) in *E. coli* and *S. cerevisiae*
- Identified proteolytic degradation, solubility and chaperone deficiency as the key bottlenecks of mass expression
- Purified enzymes via FPLC (His-tag affinity, SEC), achieving ~3.9 mg total yield for CCH11 and validating purity by SDS-PAGE and Western blot
- Characterised enzyme stability and reactivity (T_m, K_{cat}) using calorimetry and HPLC for future industrial applications

Research Intern | Jiang Research Group, Xiamen University

2024/08–2024/10

“Thiourea-Based Molecular Cages for Anion Recognition”

- Designed and synthesised tri-thiourea molecular cages for multivalent anion recognition
- Characterised structure and sulfate coordination (up to 6 ions) using NMR, CD, and SCXRD

INTERNSHIP

Shanghai Dynamax Group | Investment Analyst (Private Equity)

2024/03

Market Research and Technical Due Diligence in Biotechnology

- Conduct technical due diligence on scaffold-based 3D cell culture start-ups, developed a structured evaluation framework across 3 domains: technology principles, product performance, and commercial scalability
- Analyzed 7 competitors (e.g. matrigel, synthetic hydrogels) using 25+ scientific papers, market reports, and company data, supported by Excel-based comparative analysis
- concluded three key insights:
 - Synthetic hydrogels offer superior tunability but face cost barriers
 - Biological matrices retain translational advantages despite variability
 - Low reproducibility, complex preparation workflows, and limited throughput remain the central bottlenecks

SKILLS

Technical: Pymol, ChimeraX, MestReNova, ChemDraw, Origin, Python, R, LaTeX, Microsoft Office, etc...

Languages: Proficient in English (IELTS 8.0 with each element of 7.0+), Native Mandarin Chinese speaker